

Analysis of Trader Behavior Across Market Sentiment Regimes (Fear vs Greed)

1. Objective

The objective of this analysis is to study how trader behavior—specifically profitability, risk exposure, trading activity, and directional bias—varies across different market sentiment regimes.

By combining Bitcoin market sentiment data with historical trader-level data, this study aims to identify behavioral patterns and hidden signals that could inform smarter trading strategies.

2. Datasets Overview

2.1 Bitcoin Market Sentiment Dataset

- Granularity: Daily
- Key fields:
Date, Classification (Fear, Extreme Fear, Neutral, Greed, Extreme Greed)

For analytical clarity, sentiment labels were consolidated into three broad categories:

- **Fear** (Fear, Extreme Fear)
- **Neutral**
- **Greed** (Greed, Extreme Greed)

2.2 Historical Trader Data (Hyperliquid)

- Granularity: Trade-level
- Key fields:
Size USD, Size Tokens (trade volume)
Closed PnL (profit/loss)
Side (Buy / Sell)
Timestamp IST
Trade timestamps were converted to daily granularity and merged with market sentiment data using the trade date.

3. Methodology

1. Cleaned and validated both datasets (null checks, datatype consistency).
2. Aligned trader data with daily market sentiment using date-based merging.
3. Conducted exploratory data analysis (EDA) across four behavioral dimensions:
 - Profitability
 - Trade size (risk proxy)
 - Risk concentration
 - Directional behavior (Buy vs Sell)

4. Exploratory Data Analysis & Insights

4.1 Profitability vs Market Sentiment

Key observations:

- Median PnL is approximately zero across all sentiment regimes, indicating that most trades are breakeven.
- Average profitability is highest during **Greed**, followed closely by **Fear**, and lowest during **Neutral** periods.
- Profit volatility (PnL dispersion) is significantly higher during Fear and Greed compared to Neutral markets.

Insight:

Emotionally driven markets amplify both profit potential and downside risk, while Neutral markets exhibit more stable outcomes.

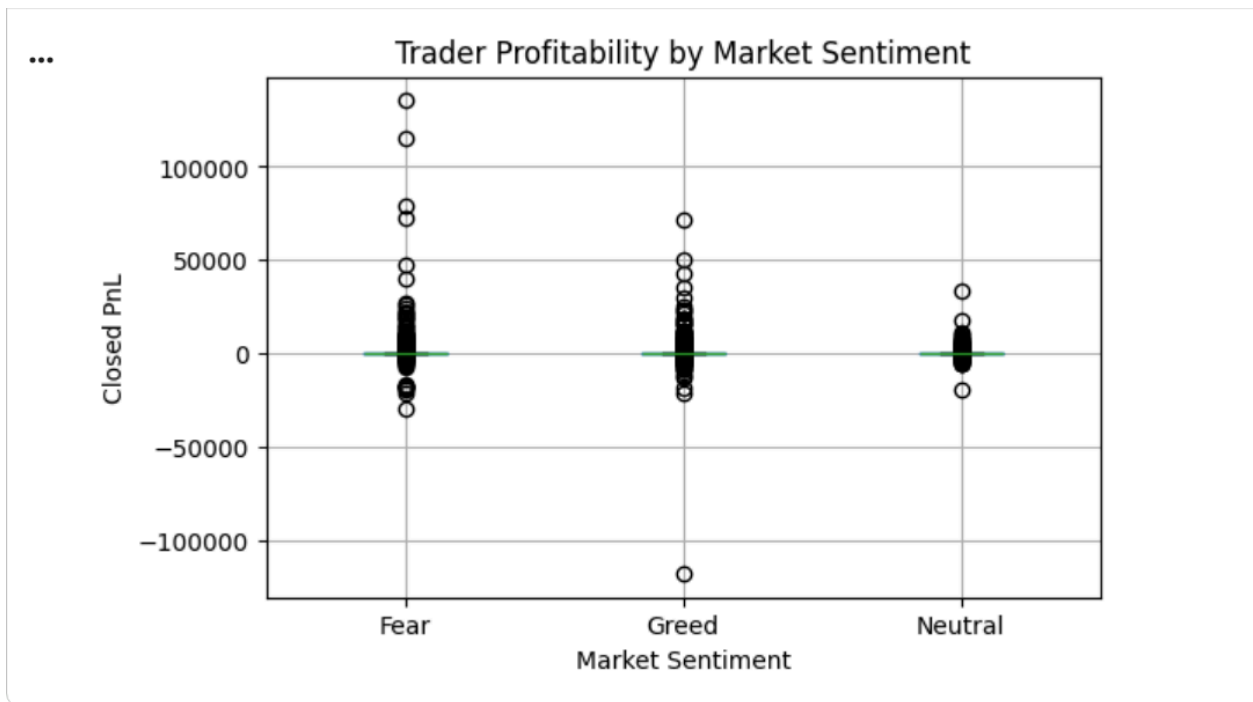


Figure 1: Trader Profitability Across Market Sentiment Regimes

This boxplot illustrates the distribution of closed profit and loss (PnL) for trades executed during Fear, Greed, and Neutral market sentiment periods. While median PnL remains close to zero across all regimes, Fear and Greed exhibit wider distributions and more extreme outcomes, indicating higher volatility during emotionally driven markets.

4.2 Trade Size (Risk Proxy) vs Market Sentiment

Trade size (USD) was used as a proxy for risk exposure.

Key observations:

- Typical trade sizes remain relatively small across all sentiments.
- Fear and Greed periods show a wider spread of trade sizes compared to Neutral periods.
- Extreme trade sizes are more frequent during Fear.

Insight:

Risk-taking behavior increases during emotionally charged markets, with Fear exhibiting more inconsistent and extreme risk exposure compared to Greed.

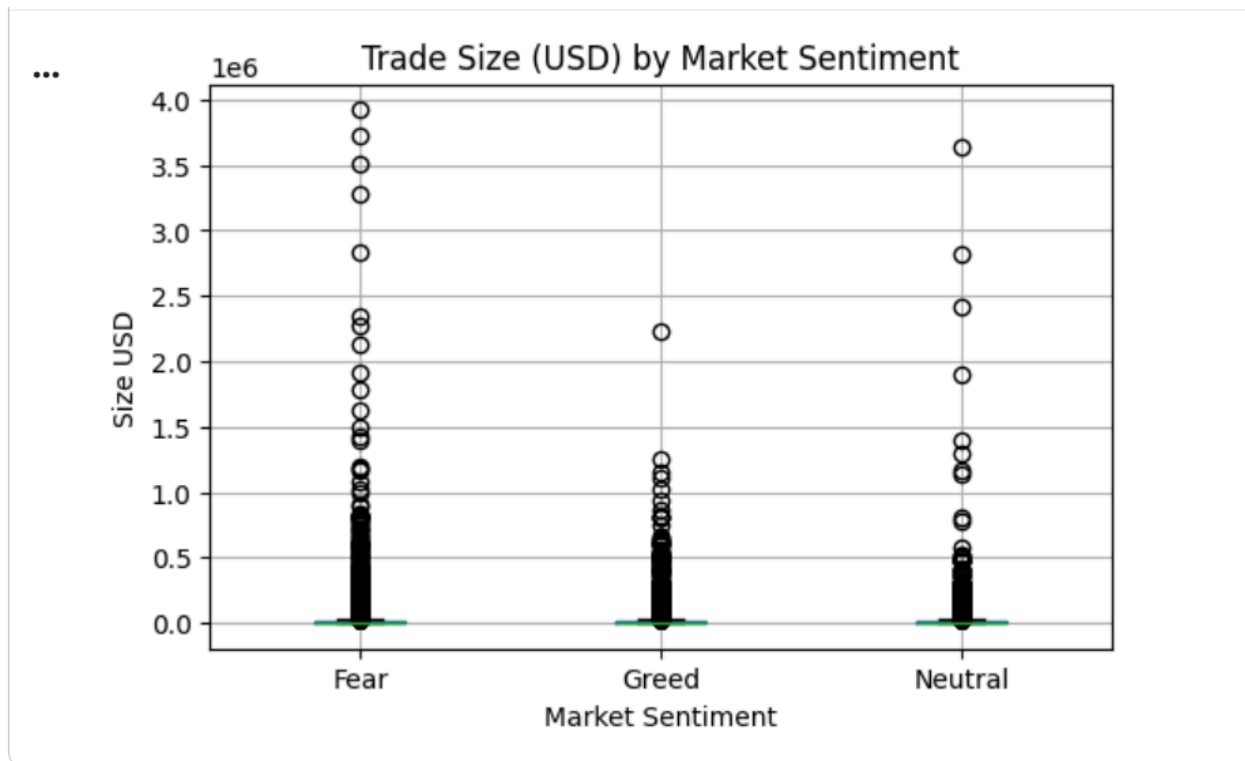


Figure 2: Trade Size (USD) Distribution by Market Sentiment

This figure shows the distribution of trade sizes in USD across different market sentiment regimes. Trade size is used as a proxy for risk exposure. Fear and Greed periods display greater dispersion and more extreme trade sizes compared to Neutral markets, highlighting increased risk-taking during emotionally charged conditions.

4.3 Risk Concentration (Quantile Analysis)

Analysis of trade size quantiles revealed:

- Median trade sizes are similar across all sentiments.
- The largest 10% of trades during Fear are significantly larger than those during Greed or Neutral periods.

- Greed shows elevated but comparatively more controlled high-end risk.

Insight:

Fear-driven markets concentrate risk in a small number of very large trades, suggesting panic-driven or forced positioning, whereas Greed reflects more systematic risk-taking.

4.4 Directional Behavior (Buy vs Sell)

Key observations:

- Fear periods show a slight **Buy bias**, suggesting contrarian or dip-buying behavior.
- Greed periods exhibit a mild **Sell bias**, consistent with profit-taking behavior.
- Neutral markets display an almost perfectly balanced Buy/Sell distribution.

Insight:

Traders tend to buy during Fear and sell during Greed, aligning with classic contrarian and profit-booking strategies.

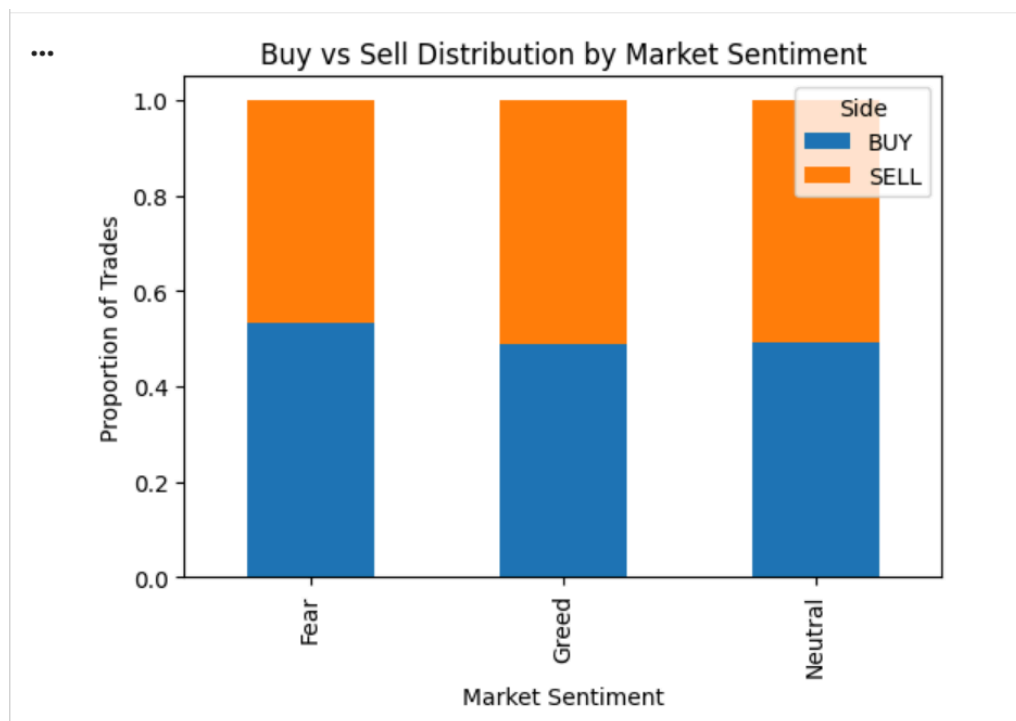


Figure 3: Buy vs Sell Trade Distribution by Market Sentiment

This stacked bar chart presents the proportion of Buy and Sell trades across Fear, Greed, and Neutral market sentiment regimes. Fear periods show a slight Buy bias, suggesting contrarian or dip-buying behavior, while Greed periods exhibit a mild Sell bias consistent with profit-taking. Neutral markets display a balanced Buy/Sell distribution.

5. Key Takeaways & Hidden Signals

- Market sentiment strongly influences trader behavior beyond price movement.
- Greed increases participation and average profitability but with controlled risk.
- Fear leads to sporadic but extreme risk exposure and higher outcome uncertainty.
- Neutral markets foster stable, routine trading behavior.
- Directional biases suggest contrarian buying during Fear and profit-taking during Greed.

6. Conclusion

This analysis demonstrates that trader behavior aligns closely with market sentiment, not just in terms of profitability but also in risk exposure and decision-making patterns.

Understanding these behavioral shifts can help traders and analysts better contextualize market movements and manage risk during emotionally driven market phases.

7. Limitations & Future Work

- Leverage data was not directly available; trade size was used as a proxy.
- Future work could incorporate:
 - Trader-level performance consistency
 - Time-lagged sentiment effects
 - Asset-specific sentiment alignment